



Efficient Computational Methods for Seismic Signal Analysis and Site Response

Tyler Davis, Prof. Chukwuebuka C. Nweke

Sonni Astani Department of Civil and Environmental Engineering, University of Southern California



ABSTRACT

Seismic site characterization is critical for earthquake hazard assessment, as local geology can dramatically amplify ground motion. The Horizontal-to-Vertical Spectral Ratio (HVSr) method offers a passive, cost-effective approach to estimating site response from ambient vibration recordings, but accurate interpretation requires time-frequency representations with sufficient resolution to capture site-specific resonance behavior. Efficient computation of Stockwell transforms and seismic data processing pipelines remain bottlenecks for large-scale geophysical analysis, with naive implementations failing to scale. Here we show that through targeted algorithmic and hardware-aware optimizations, the Stockwell Transform can be made computationally feasible for large-scale seismic recordings, enabling HVSr analysis on continuous ambient vibration datasets that were previously intractable. We developed a full processing pipeline encompassing data acquisition, quality control, high-pass filtering, and an optimized Stockwell Transform implementation. Optimizations including chunking, vectorization, GPU porting, and precision downcasting were necessary to achieve a runnable implementation on a one-and-a-half-hour continuous ambient vibration record. Kernel fusion provided a further 14% reduction in runtime, while CUDA streams introduced overhead and were excluded from the final implementation. These findings open the door to real-time or large seismic site characterization that was previously computationally infeasible, with broad implications for scalable geophysical workflows.

INTRODUCTION

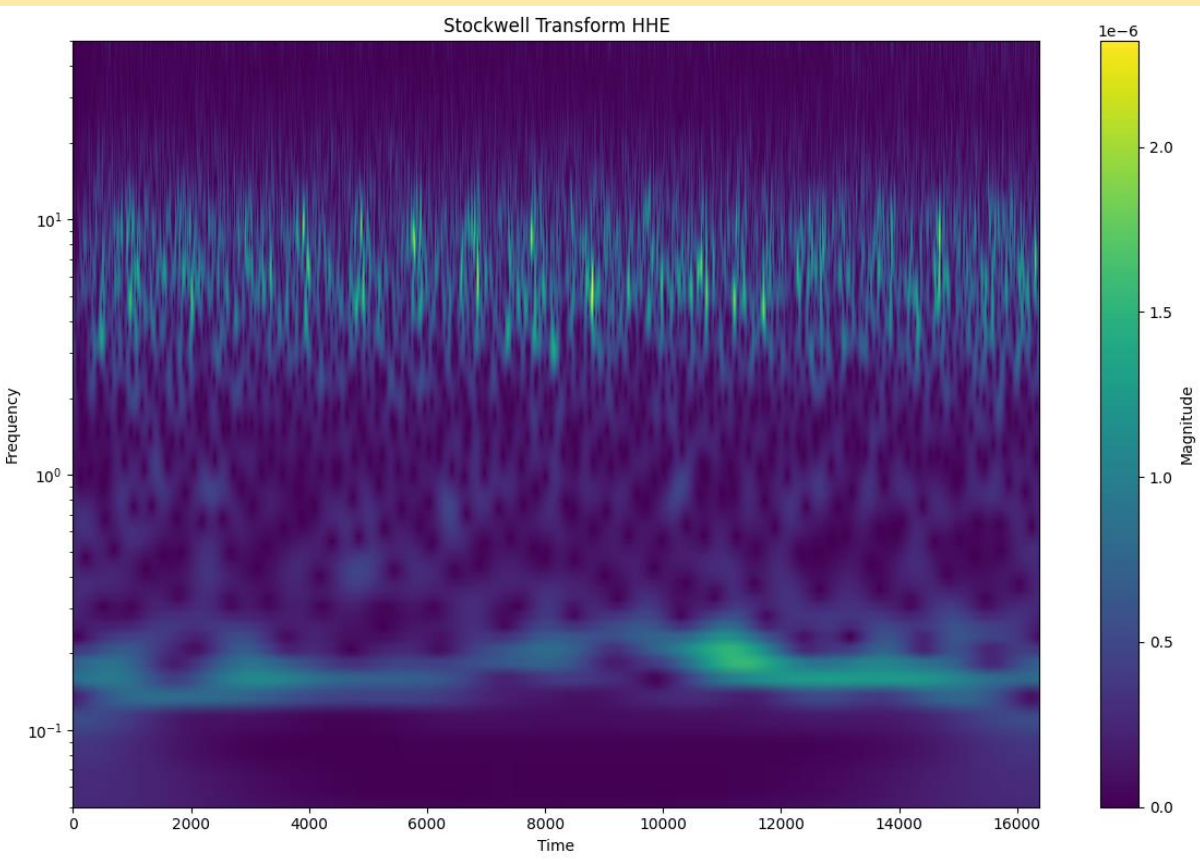
Seismic site characterization is a fundamental component of earthquake hazard assessment, as local subsurface conditions can dramatically amplify ground motion and increase structural damage. The Horizontal-to-Vertical Spectral Ratio (HVSr) method offers a passive and cost-effective approach to estimating site response by analyzing the ratio of horizontal to vertical ground motion recorded from ambient vibration. Accurate HVSr interpretation requires time-frequency representations with sufficient resolution to capture site-specific resonance behavior, making the Stockwell Transform a natural candidate due to its ability to provide frequency-dependent temporal localization. However, the computational cost of the Stockwell Transform scales poorly with record length, rendering naive implementations infeasible for the continuous, long-duration recordings required in practice. This work addresses that bottleneck through a series of algorithmic and hardware-aware optimizations, establishing a computationally viable pipeline for Stockwell-based HVSr analysis on real ambient vibration data.



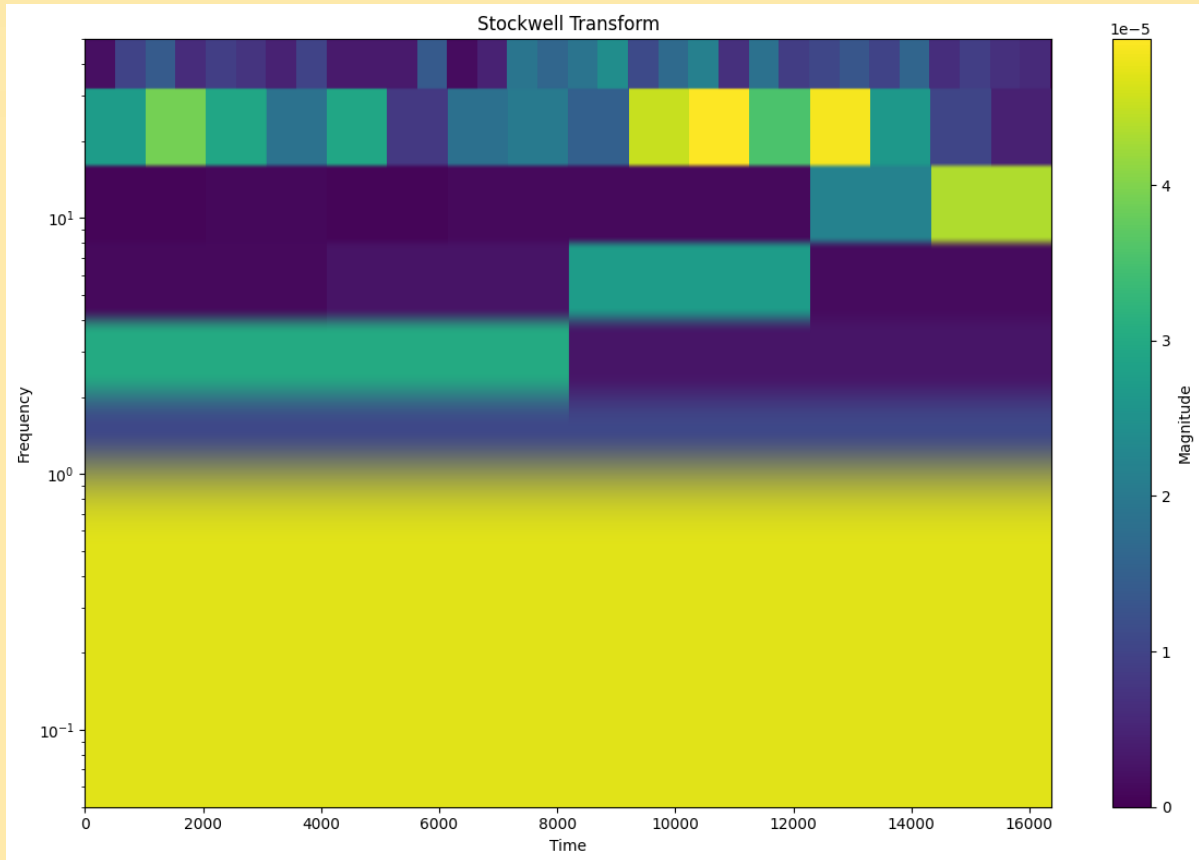
Figure 1. Consequence of earthquake shaking from the 2009 M6.3 L'Aquila Earthquake in Italy

METHODOLOGY

FDOST Investigation: The Fast Discrete Orthonormal Stockwell Transform (FDOST) is a computationally efficient variant of the Stockwell Transform and was first evaluated as a candidate for accelerating time-frequency analysis. To assess its suitability, a plotting function was developed to inspect the outputs visually. Correctness was verified through two formal checks: discrete orthonormality and Parseval's theorem, confirming energy preservation across the transform. Runtime complexity was also analytically derived and empirically validated. Despite satisfying these mathematical properties, the FDOST's time-frequency resolution proved insufficient for the demands of HVSr site response analysis, and it was not pursued further.



Stockwell Transform



FDOST

Figure 2. Depiction of the naive Stockwell Transform and the Fast Discrete Orthonormal version

METHODOLOGY (contd.)

Stockwell Transform Optimization: Given the resolution limitations of the FDOST, efforts shifted to optimizing the standard Stockwell Transform (ST). A series of progressive performance improvements were implemented and benchmarked using CuPy to quantify speedup at each stage:

- *Memory tiling (chunking):* partitioned the frequency bins into blocks of 256, ensuring the working memory footprint remained within GPU VRAM limits. Without this, the full frequency-by-time allocation exceeded available device memory and the kernel would not execute.
- *Vectorization:* replaced loop-based operations with array-level computations, eliminating per-sample overhead and enabling batch processing of frequency bins
- *Downcasting:* Reduced floating-point precision where numerically acceptable. This halved memory bandwidth requirements and improved throughput with minimal impact on accuracy.
- *GPU Porting:* the transform could be migrated to GPU execution via CuPy, exploiting massively parallel hardware to accelerate the FFT and Gaussian windowing operations.
- *Kernel Fusion:* merged the Gaussian window computation into a single fused CUDA kernel, reducing the number of memory round-trips between sequential operations.
- *CUDA streams:* explored as a means of overlapping computation and data transfer across frequency chunks. However, stream-based execution introduced synchronization overhead that outweighed any latency gains at the tested chunk sizes, and this approach was not included in the final implementation.

Data Analysis: Real seismic recordings were obtained from the Southern California Seismic Network (SCSN) via obspy retrieval package functions and preprocessed for use with the optimized ST. Outlier detection was performed using a signal drift threshold to identify and characterize anomalous signals within the dataset. The transform was then deliberately applied to outlier records to assess robustness under non-ideal conditions. A highpass filter was then applied to remove unwanted low-frequency noise, ensuring the spectral content fed into the HVSr computation reflected true site response rather than instrument drift or corrupting environmental interference (like HVAC machinery or similar).

Outlier Detection:

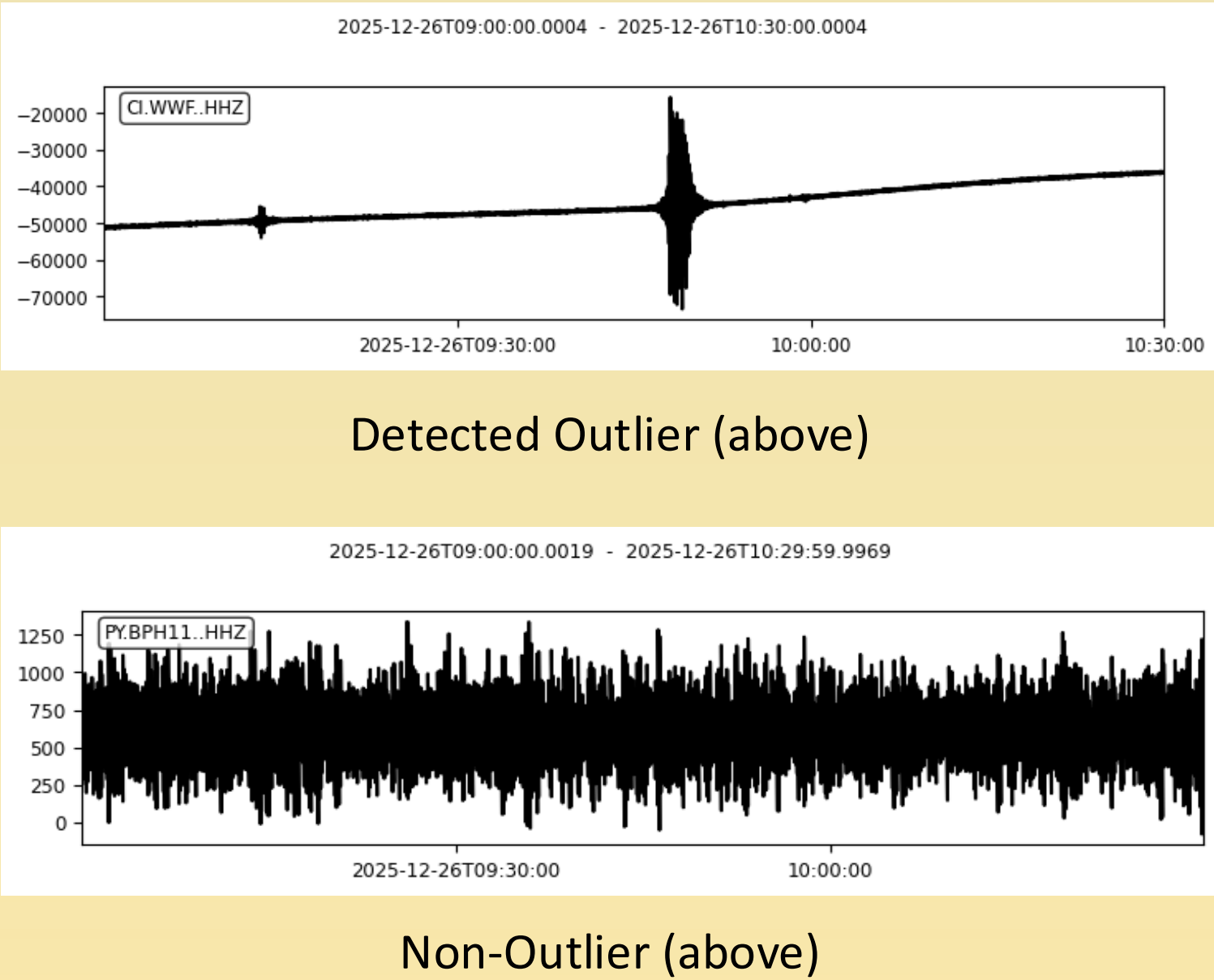
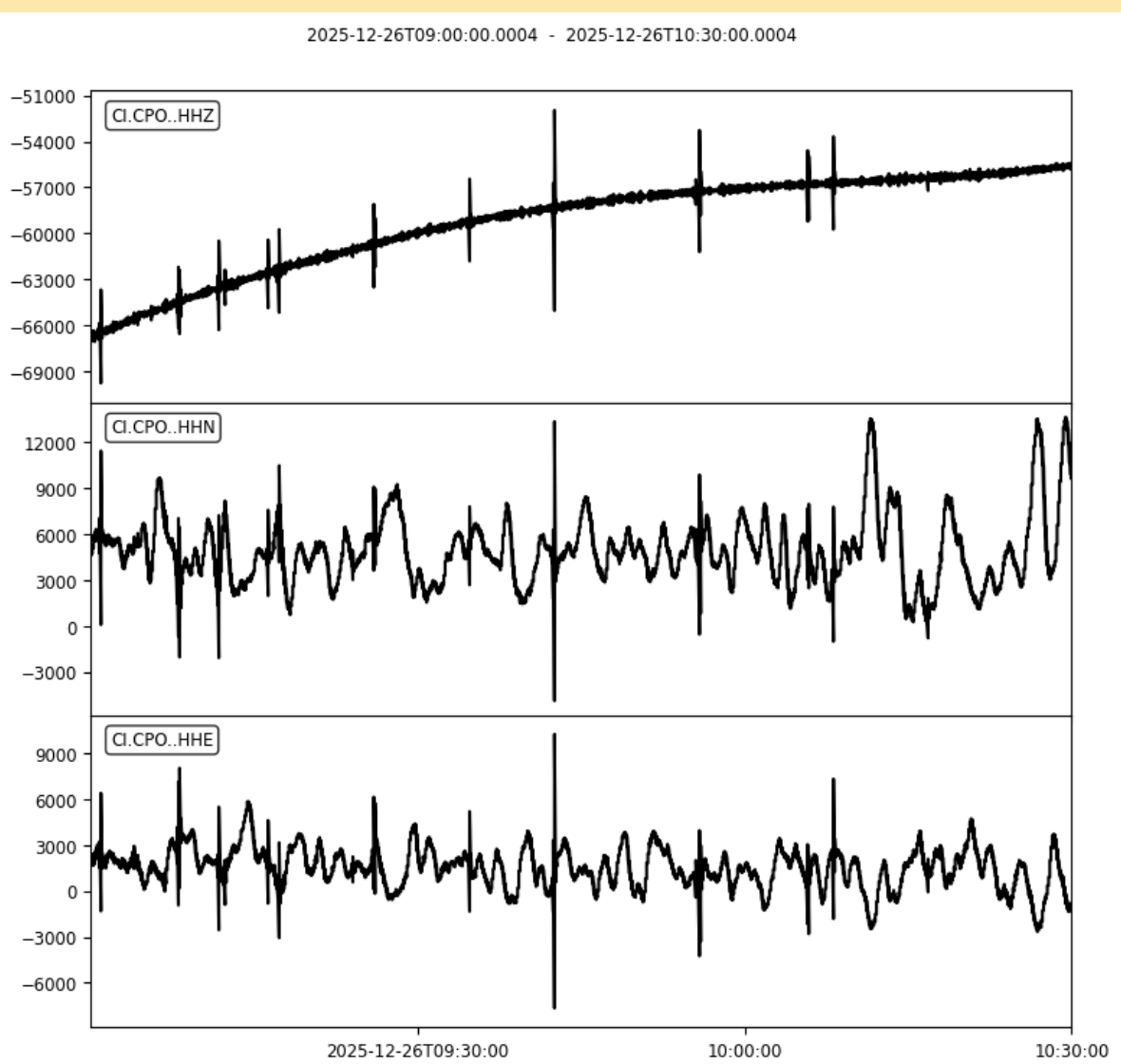
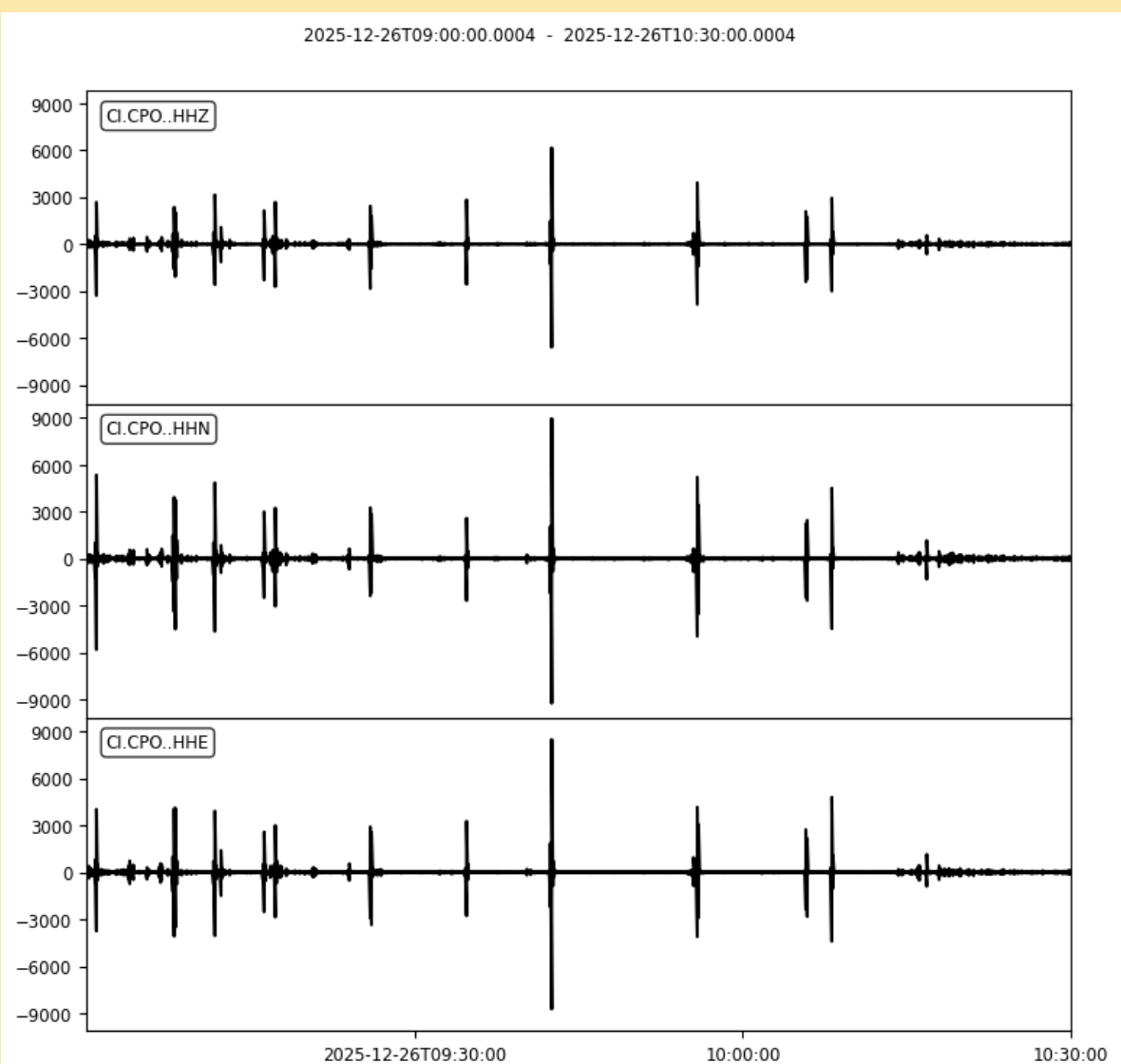


Figure 3. Examples of outlier waveforms versus non-outlier waveforms

Highpass Filter:



Before Filter



After Filter

Figure 4. Example of before and after highpass filter

RESULTS

The naive CPU implementation of the Stockwell Transform was not computationally feasible for real-world seismic recordings. For an hour and a half continuous ambient vibration record, the unoptimized transform did not complete, making it effectively impossible for practical HVSr analysis.

Initial optimizations including vectorization, downcasting, memory tiling, and GPU porting were necessary prerequisites before any meaningful benchmarking could be performed. Only after these four modifications were applied together did the transform execute successfully on the full dataset, at which point kernel fusion and CUDA streams could be evaluated as incremental improvements.

Kernel fusion yielded a measurable speedup, reducing total runtime from 3.5ms to 3.0ms, a 14% improvement. CUDA streams, however, introduced synchronization overhead that outweighed any latency gains at the tested chunk sizes, increasing total runtime to 4.8ms, a 37% slowdown relative to the fused single-stream baseline. The fused, single-stream GPU implementation was therefore selected as the final optimized version.

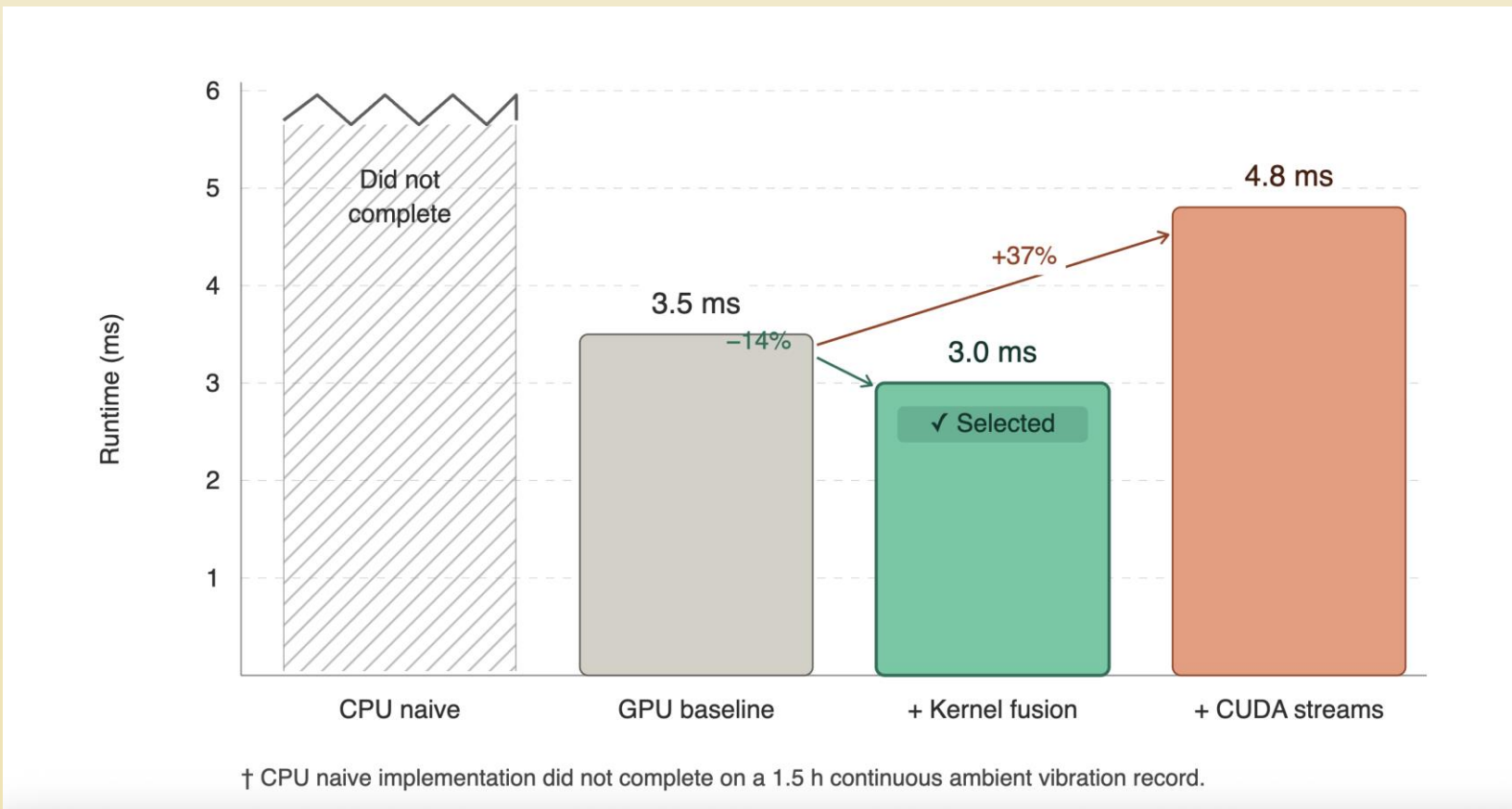


Figure 5. Runtime comparison of Stockwell Transform implementations for a 1.5-hour seismic recording

OBSERVATIONS & DISCUSSION

This work demonstrates a broader principle that extends beyond seismology: the utility of a scientifically sound method is inseparable from its computational feasibility. The Stockwell Transform has long been recognized as a theoretically attractive tool for time-frequency analysis, yet its adoption in large-scale workflows has been limited precisely because naive implementations do not scale. This tension between theoretical rigor and computational tractability is not unique to seismology. Across civil and structural engineering and geotechnical analysis, physically grounded methods are routinely underutilized because their runtime costs are prohibitive at real-world scales. The integration of hardware-aware computing into geophysical workflows is therefore not a software engineering concern but a scientific one, as it determines which questions can be asked at all.

FUTRURE WORK

The next step is to use neural networks to see if Stockwell HVSrs can be used to predict site terms.

REFERENCES

1. Wang, Y., & Orchard, J. (2009). Fast discrete orthonormal Stockwell transform. *SIAM Journal on Scientific Computing*, 31(5), 4000–4012. <https://doi.org/10.1137/080737113>
2. Stockwell, R. G. (2007). A basis for efficient representation of the S-transform. *Digital Signal Processing*, 17(1), 371–393. <https://doi.org/10.1016/j.dsp.2006.04.006>
3. Brown, R. A., & Frayne, R. (2008). A fast discrete S-transform for biomedical signal processing. *Proceedings of the 30th Annual International IEEE EMBS Conference*, 2586–2589. <https://doi.org/10.1109/IEMBS.2008.4649729>
4. Wang, Y., & Orchard, J. (2009). The discrete orthonormal Stockwell transform for image restoration. *Proceedings of the 2009 IEEE International Conference on Image Processing (ICIP)*, 2761–2764. <https://doi.org/10.1109/ICIP.2009.5414429>